

Deep Reinforcement Learning

Advanced Algorithms

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Content

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- The evolution of modern RL algorithms
- Part (II): Improving Q -learning
 - Deterministic policy gradient
 - Deep deterministic policy gradient (DDPG)
 - TD3
 - Soft Actor-Critic

Where are we?

Chapter	Topic	Content
Basics & tabular methods		
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
Deep-learning-based methods		
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	Value estimation with function approximation
8	Deep Q-learning	Q-learning with neural networks
9	Policy gradients	Direct optimization of the policy
10	Actor-critic algorithms	Improved policy gradients via value functions
11	Advanced algorithms (Part I): From policy gradient to PPO	The evolution of modern RL algorithms
12	Advanced algorithms (Part II): From Q -learning to Soft Actor-Critic	The evolution of modern RL algorithms
Advanced Topics		

The evolution of modern RL algorithms

The evolution of modern RL algorithms

(I) Policy gradient methods

- Policy is explicitly parameterized and optimized directly:

$$L(\phi) = \mathbb{E}_{\tau \sim p_{\phi}(\tau)} [r(\tau)].$$

- Gradient descent: $\phi \leftarrow \phi + \alpha \nabla_{\phi} L(\phi)$.
- Methods are typically **on-policy**.
- **Algorithms:** REINFORCE $\rightarrow$ Actor-Critic $\rightarrow$ Natural Policy Gradient $\rightarrow$ Trust-region policy optimization (TRPO) $\rightarrow$ Proximal policy optimization (PPO).

Shortcomings

- High variance gradients.
- Unstable policy updates.
- Catastrophic performance collapse.

Improvement strategy

- How to safely update policies.

(II) Value-based methods

- Approximation of the Q -function:

$$Q^*(s, a) = r + \max_{a' \in \mathcal{A}} Q^*(s', a').$$

- Extract implicit policy: $\pi(s) = \arg \max_{a \in \mathcal{A}} Q^*(s, a)$.
- Typically **off-policy**.
- **Algorithms:** Q-learning $\rightarrow$ DQN $\rightarrow$ Deep deterministic policy gradient (DDPG) $\rightarrow$ TD3 $\rightarrow$ Soft actor-critic (SAC).

Shortcomings

- Instability from bootstrapping.
- Overestimation bias.
- Maximization over actions / continuous action spaces.

Improvement strategy

- Stabilizing Q -learning with function approximation.

- Solving the continuous argmax problem.

Part (II): Improving Q -learning

Deterministic policy gradients

Deep deterministic policy gradient (DDPG)

TD3

Soft actor-critic (SAC)

References